

MTH 1220 — Calculus II

Overview

Course Information

Course	MTH 1220 -- Calculus II
Semester	AY 2026-2027 Fall
Credits	4
Days / Time	MWF 8:00-9:15am
Location	DSC 259
Prerequisites	MTH 1120 Calculus I, with a grade of C- or better.

Instructor Information

Instructor	Dr. Jacob Pichelmeyer, PhD
Email	jpichelmeyer @carthage.edu
Office	DSC 72
Office Hours	M 10:30am- 1:30pm T 11:35am-12:35am W 10:30am-11:30am

Course Description

A study of transcendental functions, infinite series, mean-value theorem, polar coordinates, integration, and applications of integration.

Student Learning Outcomes

1. Apply techniques of integration to compute reasonable antiderivatives.
2. Build connections between differentiation and integration using the Fundamental Theorem of Calculus.
3. Communicate using correct mathematical notation, coherent explanation, and appropriate form in a clear and unambiguous way, both formally and informall
4. Demonstrate sufficient knowledge of the course content
5. Explain the meaning of convergence and the ideas behind convergence tests

Instructional Methods

Lecture

Course Materials

Sisson, Szarvas, Calculus with Early Transcendentals (2nd edition), writing utensil(s), paper

Course Assignment, Assessments, and Grading Policy

Grading Policy

{'A': '93', 'A-': '90', 'B+': '87', 'B': '83', 'B-': '80', 'C+': '77', 'C': '73', 'C-': '70', 'D+': '67', 'D': '63', 'D-': '60', 'F': '0'}

Assessments

Assessment	Weight	Description
Exam 1	25%	In class, sit-down, paper exam. No notes or resources outside of a pencil and eraser. Covers the first module of the course.
Exam 2	25%	In class, sit-down, paper exam. No notes or resources outside of a pencil and eraser. Covers the second module of the course.
Exam 3	30%	In class, sit-down, paper exam. No notes or resources outside of a pencil and eraser. Covers all three modules of the course.
Micro-quizzes	5%	Given spontaneously to assess understanding of something very recently discussed. Short (5 min or less). Small stakes.
Quizzes	15%	At the end of each week, a short in-class quiz on the prior week's reading and concepts.
Total	100%	

Grading Scale

Letter Grade	Percentage Range
A	93% - 100%
A-	90% - 92%
B+	87% - 89%
B	83% - 86%
B-	80% - 82%
C+	77% - 79%

C	73% - 76%
C-	70% - 72%
D+	67% - 69%
D	63% - 66%
D-	60% - 62%
F	Below 60%

Course Policies and Procedures

Attendance Policy

Students are expected to attend class. Though no formal attendance will be taken, many days will feature a microquiz, which will act as a form of attendance for that day.

Expectations for Classroom Behavior

We all thrive when everyone feels safe, respected, and included. Bigotry, bullying, or mistreatment of any kind will not be tolerated in our classroom.

Late Work/Make-up Policy

Late work will not be accepted.

Academic Integrity Statement

It is expected that students will abide by the college's [Academic Honesty Policy](#).

Artificial Intelligence Use Policy

AI, like any technology, carries risks and benefits. It is to your advantage to understand both clearly.

Risks: heavy, *unreflective* AI use is linked to *cognitive offloading*. Research suggests that relying on AI to produce answers, rather than to support your own reasoning, can reduce engagement with the thinking process itself and may weaken independent problem-solving over time. [\[1\]](#) [\[2\]](#)

Benefits: When used as a Socratic partner, debugger, or explainer rather than an answer generator, AI has been shown to improve critical thinking, argumentation, and conceptual understanding. Students who actively critique AI output rather than accepting it at face value tend to learn more than students who simply defer to it. [\[1\]](#) [\[2\]](#)

Learning to use AI well is a valuable skill, but it cannot replace your independent reasoning, judgment, and creativity. While companies use AI to automate routine knowledge work, human value lies in the ability to critique these tools, know when to question them, and go beyond what they can produce. This class is your opportunity to develop those unique, non-automated strengths. Ultimately, your worth is not defined by economic productivity or outperforming a machine. Rather, the goal for this class is to cultivate a deep conceptual understanding and human agency that algorithms will struggle to replicate.

All work you turn in is expected to be your own. In order to earn full points on assessments, you must be able to fully and clearly explain all aspects of your work.

Student Support Resources

Student-Athletes

If you are a student athlete, you should inform me of this status at the start of the semester by providing me with the Coach's Letter that you received (this letter should name your head coach and the competition dates, and should have been provided to you at the first meeting). In addition to sharing this information at the start of the semester, you need to notify me of any class days to be missed due to a Carthage Athletics-sponsored event in which you are participating. You should notify me in advance of each upcoming contest. (For example, if you know you are traveling with the team on Friday, please let me know several days in advance). You will need to make up any missed lectures, assignments, and/or exams. Some work that can happen only during class time, such as discussion points, is not possible to make up. Students are encouraged to be in communication often; excessive absences will necessitate conversations with the instructor and the associate athletic director, and may impact your grade. Please note: student-athletes cannot miss class to attend a practice session or athletic training appointment.

Learning Accessibility Service Information

Carthage College strives to make all learning experiences as accessible as possible. If you anticipate or experience academic barriers due to your particular circumstances (including mental health, learning disorders, and chronic medical conditions), please let me know immediately so we can discuss options privately. To establish reasonable accommodations, you must register with Warren Wolchuk in Learning Accessibility Services (wwolchuk@carthage.edu).

The Health and Counseling Center

The Health and Counseling Center (HCC) addresses student physical, mental, and emotional well-being. All services, provided by experienced professionals, are free and confidential to currently enrolled, full-time undergraduate students. Students must call or visit the HCC to schedule an appointment. Health services are available by walk-in or appointment from 8:30am to 1pm and 2pm to 4:00pm. Counseling walk-in sessions are available Monday through Friday from 11:30am to 1:00pm. Last walk-in appointment begins 30 minutes before the end of walk-in hours. TWC, first floor (behind mailboxes), 262-551-5710. Students can visit Uwill to access teletherapy services.

Writing Center

The Brainard Writing Center is a free resource for student writers. The center is staffed by undergraduate Writing Fellows who have been recommended by Carthage faculty and trained to work with other students on their writing. They can work with you at all stages of the writing process, including understanding the assignment, brainstorming ideas, drafting, revising, and proofreading. Appointments are available in-person, via Zoom, or as written feedback, and can be scheduled on WOnline. For more information, visit carthage.edu/writing-center.

Peer Tutoring

Peer Tutoring is available to assist you with any aspect of the class, including understanding readings, preparing for quizzes, and studying for exams. This free campus resource is here to help everyone maximize their academic potential. Please take advantage of it. You can schedule an appointment up to seven days in advance and see the schedule on WOnline. For more information, visit carthage.edu/tutoring.

Course Specifics

Workload Expectation

For every credit this course is worth, it is expected that you will spend 2-3 hours *outside of our in-class time* on work related to the course, which includes (but is not limited to) studying lessons, working on projects, and seeking assistance from and working with the instructor during office hours. This course is worth **4 credit(s)**, so plan on roughly **8 to 12 hour(s)** of outside effort per week. This number may be much higher or lower, depending on the knowledge and experience you bring into the course.

Calendar of Course Activities

Week	Topic
Discrete sums	
Week 1	Sequences (10.1), Series (10.2)
Week 2	Comparison (10.4), Ratio/Root test (10.5)
Week 3	Absolute Convergence (10.6), Power Series (10.7)
Week 4	Taylor Series (10.8)
Week 5	Review/Flex, Exam 1
Continuous sums	
Week 6	Riemann Sums (5.1), Definite Integral (5.2), FTC/MVT (5.3), Indefinite Integral (5.4), U-Substitution (5.5)
Week 7	Integration by parts (7.1), Partial fractions (7.2)
Week 8	Trigonometric integrals (7.3), Trigonometric substitution (7.4)
Week 9	Numerical integration (7.6), Improper integrals (7.7)
Week 10	Review/Flex, Exam 2
Geometric applications	

Week 11	Volumes by slices (6.1), Volumes by shells (6.2), Arc Length (6.3)
Week 12	Parametric equations (9.1), Calculus with parametric equations (9.2)
Week 13	Polar coordinates (9.3), Calculus with polar coordinates (9.4)
Week 14	BREAK
Week 15	Flex week
Week 16	Review/Flex, Exam 3